

COUNTING

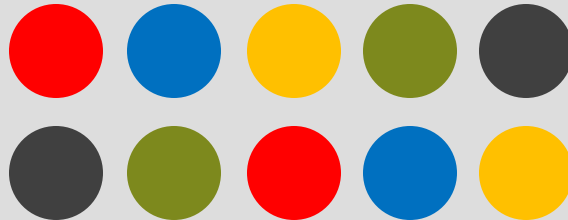
CSI 219 – Discrete Mathematics

PERMUTATION

- Permutation: In how many ways can we select r elements from a set of n elements, such that the order of the selected elements matter.
- For example: How do we choose 5 students out of 10 and **arrange them in one line?**

PERMUTATION

- **Ordered** arrangement of distinct objects



PERMUTATION

- **Ordered** arrangement of distinct objects
- r -permutation: ordered arrangement of r objects from a set of n objects
 - Number of r -permutations of a set with n elements is denoted by $P(n, r)$
 - How to count $P(n, r)$?

PERMUTATION

- How do we choose 5 students out of 10 and **arrange them in one line?**
 - The first student can be chosen in 10 ways
 - The next student in 9 ways
 - The next in 8 ways
 - The next in 7
 - The last student can be chosen in 6 ways
 - By product rule, $P(10,5) = 10 \times 9 \times 8 \times 7 \times 6 = 30240$
- We can solve permutations using the product rule.

PERMUTATION

- We can solve permutations using the product rule:
- If we want to fill r positions from a set of n elements, such that their order matters, that is, $P(n, r)$:
 - We have n choices for the 1st position
 - $n - 1$ choices for the 2nd position
 - $n - 2$ choices for the 3rd position
 - Do you see a pattern?
 - For the x^{th} position, we will have $(n - x + 1)$ choices
 - Therefore, we will have $n - r + 1$ for the last (i.e., r^{th} position)

PERMUTATION

- Therefore:

$$P(n, r) = n * (n - 1) * (n - 2) * \dots * (n - r + 1)$$

This can be written as:

$$P(n, r) = \frac{n!}{(n-r)!}$$

PERMUTATION

- What if we want to find the different orders in which **n objects can be arranged from a set of n elements.**
- That is, we want: $P(n, n)$
- This can be written as:

$$P(n, n) = \frac{n!}{(n-n)!} = \frac{n!}{0!} = n!$$

COMBINATION

- **Unordered** selection of distinct objects
- r -combination: unordered selection of r objects from a set
 - Number of r -combinations of a set with n elements $= \mathcal{C}(n, r)$
 - How to count $\mathcal{C}(n, r)$?
 - Essentially, we just need the number of subsets containing r elements, from the set of n elements

COMBINATION

- Suppose we want to select 2 students for the same position of a committee from 4 students -- $C(4, 2)$
- Consider the 4 students to be a set of 4 elements, where each element is a student:

$\{a, b, c, d\}$

- How many subsets containing 2 elements can we have?

$\{a, b\}$ $\{a, c\}$ $\{a, d\}$ $\{b, c\}$ $\{b, d\}$ $\{c, d\}$

- Note that, here, the order in which we select the elements (students) do not matter. $\{a, b\}$ is the same as $\{b, a\}$
- Therefore, $C(4, 2)$ is 6.

LET'S FIND OUT A BETTER WAY TO COMPUTE COMBINATIONS!

- Let's go back to permutation again!
- How many ways can we have r-permutations from a set of n elements?
 - Find out $P(n, r)$
- We can now solve this permutation, $P(n, r)$, a bit differently:
 - Take all the combinations of subsets containing r elements, i.e., $C(n, r)$
 - Find out the different permutations of each of the r-element subsets, that is, find out $P(r, r)$
- $P(n, r) = C(n, r) * P(r, r)$

LET'S FIND OUT A BETTER WAY TO COMPUTE COMBINATIONS!

$$P(n, r) = C(n, r) * P(r, r)$$

$$C(n, r) = \frac{P(n, r)}{P(r, r)}$$

$$C(n, r) = \frac{n!}{(n - r)!} \div r!$$

$$C(n, r) = \frac{n!}{(n - r)!r!}$$

COMBINATION

- In general, we can deduce the following:

$$C(n, r) = \frac{P(n, r)}{r!}$$

$$C(n, r) = \frac{P(n, r)}{r!} = \frac{n!}{(n-r)!} \times \frac{1}{r!} = \frac{n!}{r!(n-r)!}$$

EXAMPLE

How many ways to award gold, silver and bronze medals from 8 runners in a race?

$$\text{Solution: } P(8,3) = 8 \times 7 \times 6 = 336$$

Note: There must not be a tie!!!

EXAMPLE

How many permutations of the letters $ABCDEFGH$ contain the string ABC ?

Solution: Considering ABC as one object, there are 6 objects in total.

Thus, total # of permutations = $6! = 720$

EXAMPLE

How many poker hands of five cards can be dealt from a standard deck of 52 cards?

Solution: Order of the five cards is not important, so the problem reduces to choosing 5 cards from 52.

$$\text{Thus, } C(52,5) = \frac{52 \times 51 \times 50 \times 49 \times 48}{5!} = 2598960$$

EXAMPLE

How many bit strings of length n contain exactly r 1s?

Solution: We have to choose r positions for the 1s among the n available positions.

Thus, the # of bit strings is $C(n, r)$

EXAMPLE

How many bit strings of length 10 contain

- a) exactly four 1s?
- b) at most four 1s?
- c) at least four 1s?

EXERCISE

There are 9 Math faculties and 11 CS faculties in a university. The authority wants to form a committee to develop a Discrete Math course, such that 3 members of the committee are Math faculties, and 4 are CS faculties. The members will all be in the same position

How many ways to form the committee?